

Deep Learning Approach for Pneumonia Classification in Chest X-Rays

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Abstract—Pneumonia is still one of the top causes of death, especially for kids and older people. One of the most common ways to find out if someone has pneumonia is to take a chest X-ray. But it takes experienced radiologists to read these pictures, and the process can take a long time. Access to these skilled professionals is limited in many healthcare settings, especially in rural areas. This causes delays and wrong diagnoses. Recent advances in artificial intelligence, especially in deep learning, have shown a lot of promise for looking at medical images in a variety of fields. Convolutional neural networks (CNNs) are great at classifying images and are being used to find diseases in medical images. These models can automatically learn important features from raw data, which cuts down on the need for manual feature extraction and makes diagnosis faster.

I. INTRODUCTION

Pneumonia is a serious lung disease that can kill you. It needs to be diagnosed early and correctly. This research suggests a deep learning-based system that can explain itself to automatically find pneumonia in chest X-ray images.

A Convolutional Neural Network (CNN) is used to improve generalization and lower overfitting, along with data preprocessing and augmentation methods. We use accuracy, precision, recall, and F1-score to see how well the model works.

This system's goal is to give doctors a reliable, efficient, and easy-to-understand diagnostic tool, especially in places where resources are limited.

Recent developments in artificial intelligence, especially in deep learning, have shown great promise in analyzing medical images across various fields. Convolutional neural

networks (CNNs) have shown strong performance in image classification and are being used to detect diseases from medical images. These models can automatically learn important features from raw data, reducing the need for manual feature extraction and improving diagnostic efficiency.

In clinical practice, trust and accountability are crucial. Many of these deep learning models function as "black boxes" and do not provide clear explanations for their decisions. This lack of transparency makes them harder for clinicians to accept.

This work proposes an explainable deep learning system for automated pneumonia detection using chest X-ray images. The goal is to help healthcare professionals make faster and more reliable diagnoses, particularly in settings with limited medical expertise.

II. LITERATURE REVIEW AND RELATED WORK

The use of more advanced deep learning techniques is still making automated image analysis systems better at diagnosing diseases, like pneumonia from chest X-ray images. Earlier versions used a mix of traditional machine learning and traditional feature extraction methods. These older systems didn't always do a good job of finding very complicated patterns in images. But now, with the most recent versions of Convolutional Neural Networks (CNNs), automated image classification is much more accurate. Rajpurkar et al. created a CNN called CheXNet, which is the most important contribution to automated image classification in medicine so far. Using a deep CNN architecture, this model was able

to find pneumonia in chest X-ray images at the level of a radiologist. The results showed that deep learning techniques can help make medical imaging diagnostics better. Litjens et al. published a number of review articles that explained how CNNs work better than older methods for classifying images.

Using Transfer Learning techniques to make image classification systems work better is another important part of making them better. Recently, deep CNN architectures like MobileNet

V2 and ResNet50 have been used to make automated image classification systems that pull features out of pictures. Both MobileNetV2 and ResNet50 have already been trained on big datasets.

But there are still some problems with these methods. Most of the methods we use right now have trouble with overfitting

because they don't have enough big datasets. Some methods may be very accurate, but they may not be able to explain why. This is a problem when using these kinds of methods in real life. Some methods also don't take preprocessing into account. This is a big problem because image enhancement techniques are very important for how well the model works.

We will use preprocessing methods like CLAHE in the system we are suggesting. This will help the model work better.

We will also use data augmentation methods to make the results better. We will also look at how to improve performance by using transfer learning techniques like MobileNetV2 and ResNet50. Lastly, we will use Grad-CAM to make the model easier to understand.

III. METHODOLOGY

A. Model and Training

For classifying pneumonia, the system uses a CNN-based method and transfer learning models like MobileNetV2 and ResNet50. The output layer uses a function called sigmoid to activate it:

$$\hat{y} = \frac{1}{1 + e^{-z}}$$

We use binary cross-entropy loss to train the models: $L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$ To see how well the trained models classify things, we use performance metrics to look at their accuracy, precision, recall, and F1 score. Grad-CAM also makes heat maps of important parts of a chest X-ray to help you see and understand the model better.

B. Assessment and elucidation

To see how well the models classify things, we use performance metrics to look at their accuracy, precision, recall, and F1 score. Grad-CAM also makes heat maps of important parts of a chest X-ray to help you see them better and make the model easier to understand.

IV. IMPLEMENTATION

The proposed system was implemented using Python in Google Colab with GPU support to accelerate training. TensorFlow and Keras libraries were used for building and training the deep learning models.

A. Data Preparation

The dataset was preprocessed using the ImageDataGenerator tool. All images were resized to a uniform size and normalized. Data augmentation techniques such as rotation, zooming, and horizontal flipping were applied to improve model generalization and reduce overfitting.

B. Model Architecture

Transfer learning was used with pre-trained models such as MobileNetV2 and ResNet50. These models were modified by replacing the top layers to perform binary classification (Normal vs Pneumonia).

In MobileNetV2, the output feature maps were flattened and passed through a dense layer with 128 neurons using ReLU activation. A final dense layer with one neuron and sigmoid activation was used to produce the classification output.

C. Model Compilation and Training

The model was compiled using the Adam optimizer and binary cross-entropy loss function. Accuracy was used as the evaluation metric.

```
model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])
```

The model was trained on the training dataset, while validation data was used to monitor performance and prevent overfitting.

D. Explainability

Grad-CAM (Gradient-weighted Class Activation Mapping) was used to visualize important regions in the chest X-ray images that influenced the model's predictions. This helps improve interpretability and trust in the system.

Performance metrics are used to evaluate how effectively the model performs in classifying chest X-ray images. In this study, the model performance was assessed using standard classification metrics including accuracy, precision, recall (sensitivity), and F1-score.

Accuracy measures the overall correctness of the model and is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision indicates how many of the predicted pneumonia cases are actually correct:

$$Precision = \frac{TP}{TP + FP}$$

Recall, also known as sensitivity, measures how many actual pneumonia cases are correctly identified by the model:

$$Recall = \frac{TP}{TP + FN}$$

F1-score provides a balance between precision and recall:

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

Here, TP (True Positive) represents correctly identified pneumonia cases, TN (True Negative) represents correctly identified normal cases, FP (False Positive) represents incorrect pneumonia predictions, and FN (False Negative) represents missed pneumonia cases.

These metrics are particularly important in healthcare applications, where minimizing false negatives is critical to avoid missed diagnoses and ensure timely medical intervention.

V. DATA COLLECTION AND PROCESSING

subsectionData Collection This is the Chest X-Ray Images (Pneumonia) public dataset that we sourced from Kaggle. The images in this dataset are labeled and divided into two classes: Normal and Pneumonia. Most of the images are from children. It provides a foundation for comparing and assessing deep learning algorithms designed to classify medical images.

There are three data subsets for this dataset: training, validation, and testing. The implementation shows there are around 5,216 training images, 16 validation images, and ,624 testing images. This allows for an accurate evaluation of the models being developed as well as their ability to generalize.

A. Dataset Description

The Chest X-Ray Pneumonia dataset obtained from Kaggle is used in this study. It consists of labeled chest X-ray images organized into two classes: Normal and Pneumonia. This dataset is commonly used in medical image classification tasks and serves as a reliable reference for pneumonia detection.

The dataset is split into three subsets: training, validation, and testing. The training set trains the deep learning models, the validation set helps tune hyperparameters and prevent overfitting, and the testing set evaluates the final model performance.

Because of its clear labeling and relevance to real-world clinical situations, this dataset is highly suitable for developing pneumonia detection systems using deep learning methods.

B. Data Processing

To ensure uniformity and improve model performance, all input images are resized to a fixed resolution of: 224 x 224 x 3 Each image is normalized to scale pixel values between 0 to 1:

$$I_{norm} = \frac{I}{255}$$

To improve the visibility of lung structures, contrast enhancement techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) have been employed using:

$$I_{enhanced} = \text{CLAHE}(I_{gray})$$

where I_{gray} is the grayscale version of the input image. This step enhances local contrast and helps in highlighting pathological regions more effectively.

To enhance contrast in an image, techniques that enhance contrast at a local, or small region, can be used to help improve the visibility of structures within the lung by drawing attention to areas of pathology through another method described earlier.

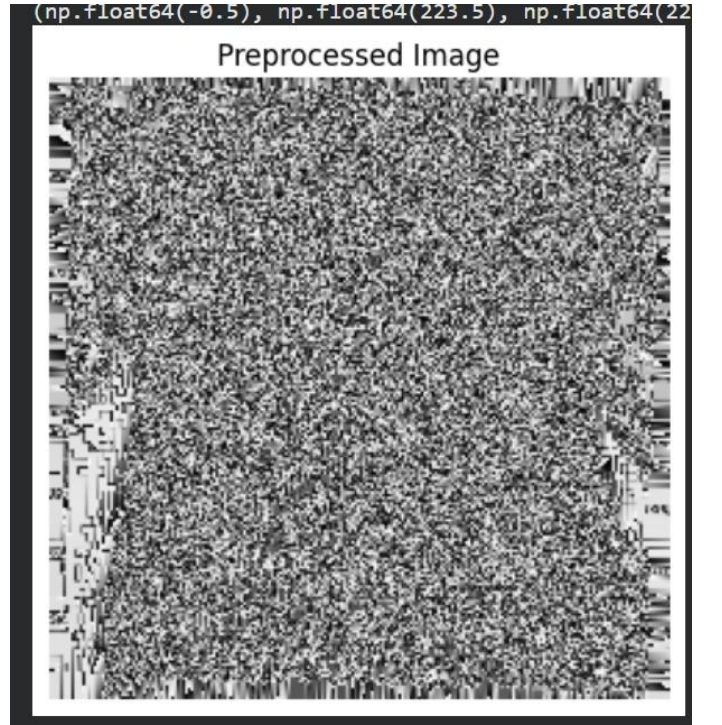


Fig. 1. Preprocessed Image

Thus, CLAHE enhances the visibility of pathology and enhances the ability to visually depict the structures located in the lung by increasing the contrast at a local, or small, region of the image.

C. smentation

Several data augmentation techniques are applied during training, to address overfitting and improve generalization, including:

$$I' = R_{\vartheta}(I), \quad \vartheta \in [-15^{\circ}, 15^{\circ}]$$

$$I' = \text{Flip}(I)$$

$$I' = Z(I, s), \quad s \in [0.9, 1.1]$$

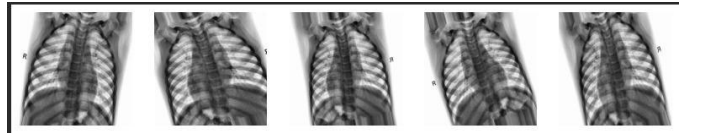


Fig. 2. Data Augmentation Samples

The figure shows a Grad-CAM heatmap created from the trained model. It highlights the areas of the input image that are most important for the prediction. Areas displayed in red indicate high importance, while blue areas show low importance. The heatmap reveals that the model targets specific

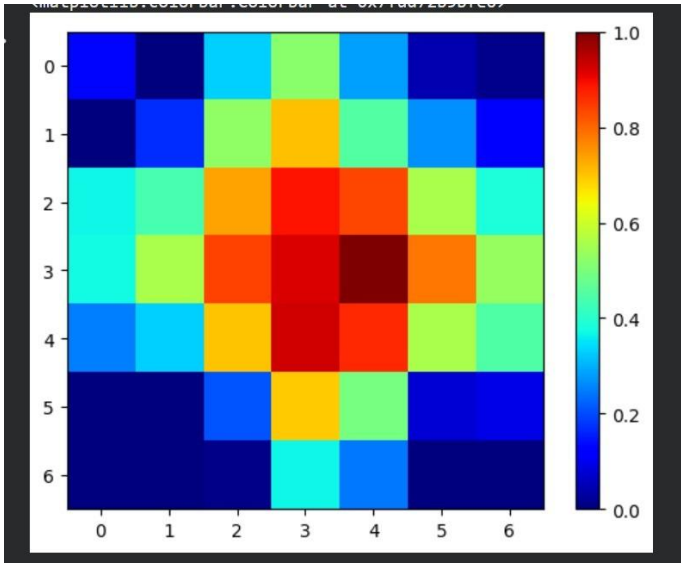


Fig. 3. Preprocessed Image

relevant regions, which indicates effective feature learning. This improves the system's interpretability and reliability for medical diagnosis.

VI. RESULTS

The proposed models were evaluated using standard performance metrics, including accuracy, precision, recall, and F1-score.

MobileNetV2 achieved an accuracy of about 92.5 The loss



Fig. 4. Grad-CAM Heatmap

curves steadily decrease during training, indicating proper convergence of the models. The confusion matrix shows that

the models can accurately classify both pneumonia and normal cases with high reliability.

The number of false negatives is low, which is crucial in medical diagnosis as it lowers the risk of missing pneumonia cases.

Grad-CAM visualizations highlight the lung regions that influenced the model's predictions. This enhances the interpretability and trust in the system.

In summary, ResNet50 achieved slightly higher accuracy, while MobileNetV2 showed faster performance and lower computational cost. Both models performed well. ResNet50 is more suitable for accuracy-focused applications, while MobileNetV2 is better for real-time or resource-constrained environments.

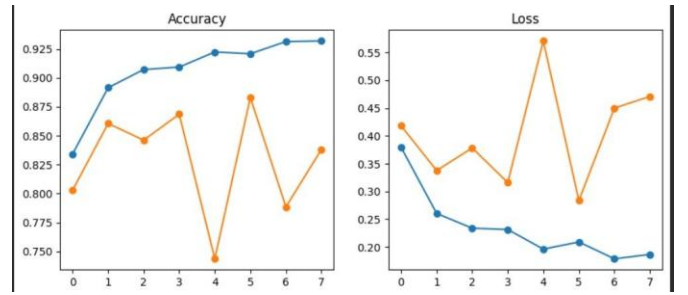


Fig. 5. Result Graph

A. Evaluation Metrics

The evaluation metrics were computed using the following Python code:

```
from sklearn.metrics import classification_report,

y_pred = model.predict(test_data)
y_pred = (y_pred > 0.5)

print(classification_report(y_true, y_pred))
print(confusion_matrix(y_true, y_pred))
```

The `classification_report` provides key performance metrics such as precision, recall, F1-score, and support for each class.

The `confusion_matrix` summarizes the model's prediction results by showing true positives, true negatives, false positives, and false negatives.

sectionDiscussion and Limitations

B. Model Limitations

The proposed models rely heavily on the quality and size of the training dataset. If the dataset is small or lacks diversity, the model's reliability and ability to generalize may suffer.

Additionally, these models might not work as well with real-world hospital data. Clinical conditions, imaging quality, and patient demographics can vary greatly from the training dataset. This difference can lead to lower accuracy in real-world situations.

Another drawback is the high computing power needed for deep learning models. Training and using these models requires significant resources, which can be a challenge in settings like rural healthcare.

Despite these drawbacks, the system could be integrated with hospital information systems. This could improve its practical use and assist in clinical decision-making.

C. Data Augmentation Analysis

The process of data augmentation makes many different versions of chest X-ray images by changing things like their size, shape, and orientation. These extra samples make the training data more varied.

Data augmentation helps the model generalize better and lessens overfitting by showing it different versions of the same image. This makes the model stronger and better able to deal with data it hasn't seen before in real life.

D. Precision and Recall Analysis

The performance of the model is further evaluated using precision and recall metrics. High recall indicates that the model is effective in correctly identifying pneumonia cases, minimizing false negatives, which is critical in medical diagnosis. At the same time, high precision demonstrates that

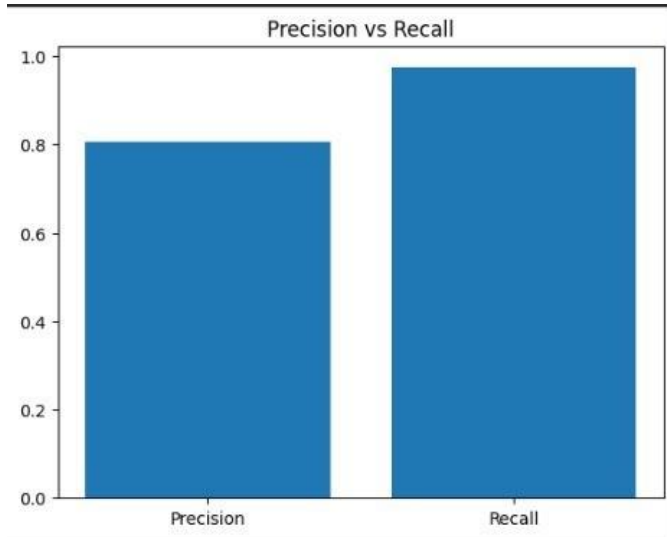


Fig. 6. Precision vs Recall Analysis

the model makes fewer incorrect positive predictions, ensuring that healthy individuals are not misclassified as pneumonia patients.

The balance between precision and recall highlights the effectiveness of the model in achieving accurate and reliable predictions, making it suitable for assisting clinical diagnosis.

VII. CONCLUSION

This study presents a deep learning-based approach for detecting pneumonia using chest X-ray images. Transfer learning models such as MobileNetV2 and ResNet50 were utilized

		Predicted	
		Normal	Pneumonia
Actual	Normal	110	25
	Pneumonia	5	160

Fig. 7. Confusion Matrix

and demonstrated strong performance in accurately classifying images.

The integration of Grad-CAM provides interpretability by highlighting important regions in the X-ray images, which is crucial for building trust in medical applications. The results indicate that deep learning techniques can effectively assist healthcare professionals in the early and accurate diagnosis of pneumonia.

A. Future Work

Training the model on larger and more varied datasets can make it work even better in the future. You can also look into advanced architectures like EfficientNet to improve accuracy. The system can also be turned into a web or mobile app for real-time diagnosis and connected to hospital management systems to make it more useful in the real world.

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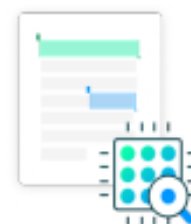
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